

Lecture
Overview

Introduction

History of
Game Spaces

Essential
Skills

Level Design
Theory



Level Design Lecture

FH Salzburg

Thomas Bauböck WS 2025

01

Lecture Overview

What is going to happen today?

Lecture Overview

- Introduction
- Level Design Theory
- Level Design Tools & Glossar
- Single Player Map Design
- Multiplayer Map Design
- Level Design: “Creating a Map for WoT” Docu
- We create a Macro Design...



02

Introduction

The humble beginning...

Expectation of the lecture?

- **Spark interest** in Level Design
- Give a first introduction to Level Design
 - **Pillars** of Level Design
 - **Terminology**
 - How to **plan a level** for a game
- What is beyond this lecture?
 - How to make a good design for a level
 - Many theories and principles
 - Practice, experience & game dependent
- If there is more interest in LD (which I hope)
 - see appendix

Who am I?

- Thomas Bauböck “zumpi”
- ~1999 started building levels
 - multiple Half Life 1 modding projects
- 2003 “Informations- Design” @ FH Joanneum
- 2006 intern semester @ Bongfish GmbH
- 2007 Master “Interaction & Media Design”
- 19+ years working at Bongfish GmbH
 - 10++ years “Level Design Department” lead
 - World of Tanks and side projects
 - Game, World & Level Designer
 - for various projects at Bongfish
- Level Design Expert - Role



What is a Level Designer

- Level Design **IS NOT** Level Art /SetDressing
- Level Design is **multi disciplinary**
 - generating more than map geometry
- Level Designers are **Game Designer**
 - integrating game mechanics
 - gameplay loops and story beats
- Level Designers understand **Level Art**
 - What does Art need to convey their goals?
- Level Designers know about **Tech Art, Sound, Animation, Code, UI, UX**
- Level Design finds solutions

*“Level Designer are **Game Designer** who **specialize** in generating the **game space** in which the players experience the game.”*

*“Level Design **is heavily influenced** by a multitude of **disciplines** in game development.
Level Design **heavily impacts** a multitude of **other disciplines** in game development.*

COLLABORATE!!



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Questions so far?



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03

History of Game Spaces

Finding the origins...

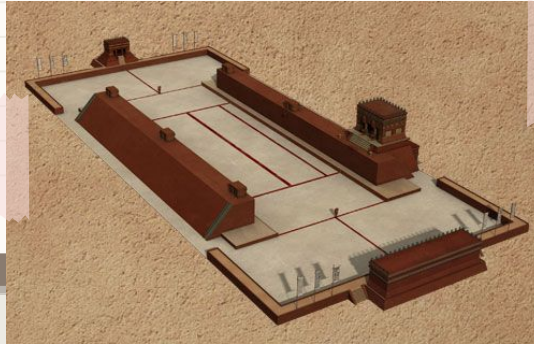
Game Spaces

- oldest known ruled game spaces

- Tsu Chu
- ~ 3000 bc.



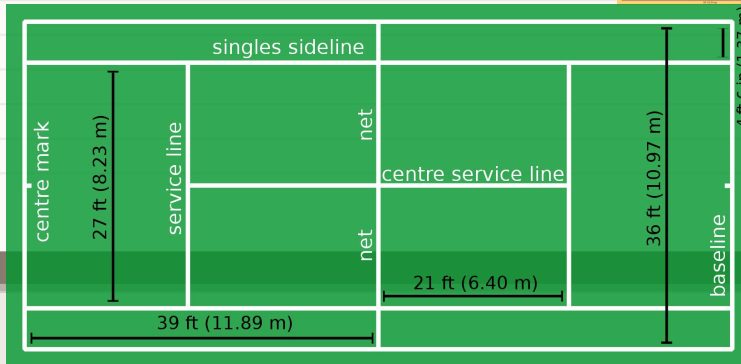
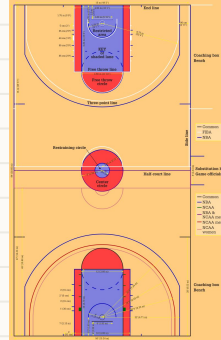
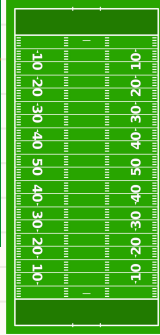
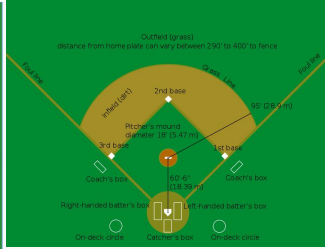
- ball playing places in mesoamerica



“playing games is a cultural good of humankind”

Game Spaces

- game spaces nowadays



Football, Baseball,
American Football,
Basketball, Tennis

Game Spaces

- electronic game spaces  Oldest “electronic computed” game space you can think of?
 - “Noughts and Crosses”/”TicTacToe” (1952)



Game Spaces

- 1958 “Tennis for Two”
 - new game space
 - oscilloscope
 - ruleset to game space
 - static game space



Game Spaces

<https://www.youtube.com/watch?v=Rmnb4Hktv7U>

- 1962 “Spacewar!”
 - developed by MIT students
 - two spaceships fight around a gravity field in the center
 - players can warp around the edge
 - extended former limited game space

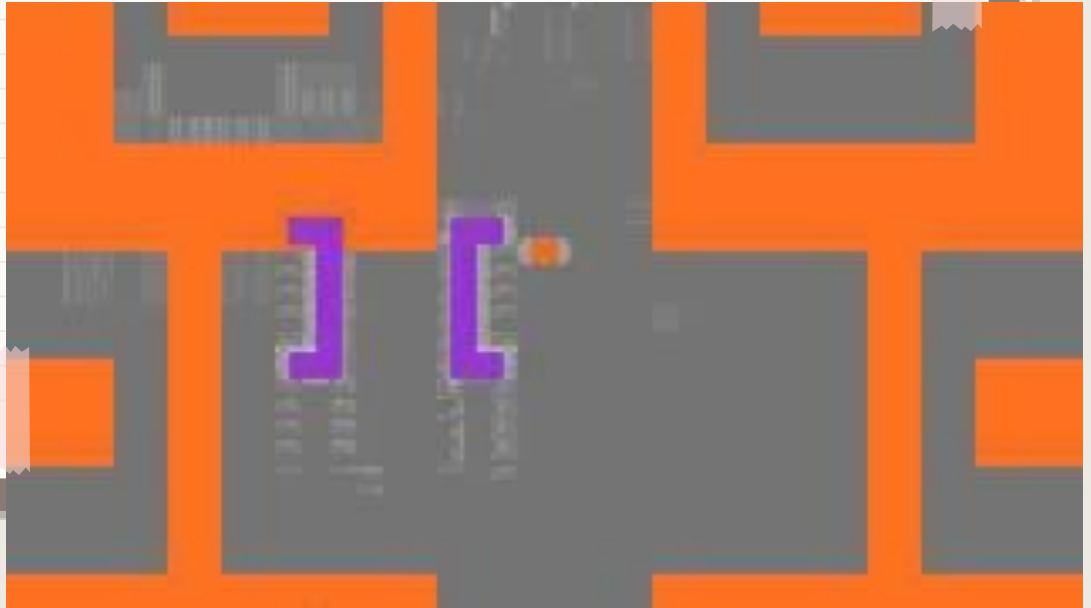


Game Spaces

- early game spaces occurred only on one screen  Any ideas why?
 - technical limitation - computing power
- game designs accommodate to these restrictions
 - short gameplay loops
 - limited game space
- push boundaries with new tech developments

Game Spaces

- Atari “Adventure” 1978
 - moving from screen to screen
 - generating larger game spaces
 - graphical standard for long time



Game Spaces

- Atari “Defender” 1981
 - continuous scrolling world



Game Spaces

- A completely different approach!



Which other medium can you think of, describing large worlds?

Game Spaces

- A completely different approach!
- 1976 Colossal Cave Adventure



Which other medium can you think of, describing large worlds?

```
In Narrow Corridor  
scrawled the inscription, "Fee fie foe  
foo" [sic].
```

```
>s
```

```
In Narrow Corridor  
You are in a long, narrow corridor  
stretching out of sight to the west. At  
the eastern end is a hole through which  
you can see a profusion of leaves.
```

```
>n
```

```
You can't go that way.
```

```
>w
```

Game Spaces

- A completely different approach!
- 1976 Colossal Cave Adventure
- 1981 Zork I

West of House

Score: 0

Moves: 0

ZORK I: The Great Underground Empire
Copyright (c) 1981, 1982, 1983 Infocom, Inc. All rights reserved.
ZORK is a registered trademark of Infocom, Inc.
Revision 88 / Serial number 840726

West of House

You are standing in an open field west of a white house, with a boarded front door.

There is a small mailbox here.

Game Spaces

- Rapid improvements in tech
- Super Mario Bros.
 - side scrolling from “Defender”
 - adventure style rooms
 - mixing with more colors
 - more challenging game play



Game Spaces

- 3D games starting simultaneously
 - 1980 “Battlezone - 3D vector world
 - other games isometric, axonometric projections as perspectival views
-
- other games isometric, axonometric projections (perspectival views)

BATTLEZONE



F1 KEY STARTS GAME

Game Spaces

- 1992 id Software
 - “Wolfenstein 3D”
 - 300x200 pixel resolution
 - one of the first “full 3D games”
 - first “Ego Shooter”
- since 2022 it is deleted from
“The list” and is rated USK 16

Game Spaces

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Led to more complex games and environments.



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Questions?

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04

“Essential” Skill set

(not only) for a Level Designer

01 - Research

- Which **environment** is the level happening in?
 - Which elements & features can be helpful or hindering?
 - Gather references, photos, concept art, mood boards, etc.
 - Does the architecture & geology support the plan/idea/task?
 - Which mood do I want to establish in this environment?
- **Document** your research!
 - The longer you work (as Level Designer), the bigger your **catalogue** grows. -> very useful also for other disciplines!

02 - Communication

- **Communicate** with other departments!
 - Do I **stay in line** with the games **vision**?
 - Does the level I am planning require any **special gimmicks**?
 - How can they be realized?
 - **Get feedback** on your ideas!
 - Take other opinions into account!
 - Are there other problems rising you haven't thought of before?

03 - Space assessment

- How much **space** do certain actions need?
 - Is the game **diverting from reality**?
 - How much **space** is **needed for** the player to perform the **planned actions**?
- Creating a **game** and not copy the real world
 - being able to know/judge volumes, distances and amounts is very helpful
 - **bending** real world **rulesets** to our needs is important
 - derive **games metrics** from those rulesets and keep them in mind!

04 - Multidisciplinary

- A **designer need to know** about all specifics
 - knowledge in other fields
 - Game Design intentions (Game loops, Gameplay Beats etc.)
 - Art Process
 - Engine Tech
 - Scripting / Code
 - UI / UX
 - Sound
- A **good level design** takes all those aspects into account and **finds synergies**, good compromise or intentional “break out” of conventions.

05 - Psychology



Why do humans like playing games?

- Games **offer** fun and entertainment
 - a healthy dose of **escapism**
 - **forget** about **stress** in the real world and refresh the brain
 - online games **offer a social aspect**
 - players are **rewarded** in a game for achieving certain tasks
- It is an **entertainment** medium
- Players should be **encouraged** to test their **skills, explore and enjoy**
 - Level Designers job to get them to do so
 - Player guidance, difficulty curves, puzzles, pacing, excitement graphs
 - > tools to get the **majority** of players do what **we** want them to experience

06 - Self awareness

- Level Design is **not easy** and needs a lot of brain processing power
 - When are you most productive?
 - Plan your day around the focus hours
 - take enough breaks
- **Reflect** over your designs
 - turn your work upside down
 - be honest with yourself & accept feedback
- *The perfect Design is an illusion.*
 - **compromises** & semi optimal decisions are daily business
 - BUT please find **GOOD compromises**
 - THINK enough before a **decision**
 - **experience** creates “gut feelings” which can drive the thought process



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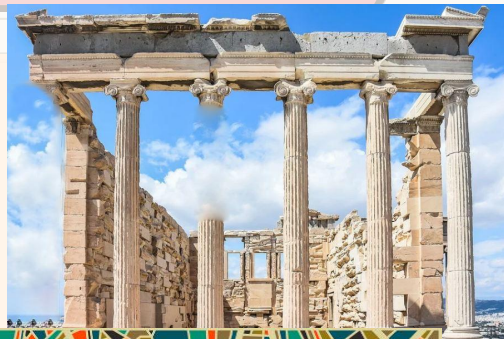
05

Level Design Theory

Real world Architecture

Real World Architecture

- TLDR:
architectural principles are important
- structural elements
How are buildings built and why?
 - Support columns
 - Spatial structures
 - squares
 - roads
 - buildings
 - and how are they connected?



Real World Architecture

- Architecture can transport **feelings**
 - security, with heavy fortified structures
 - light open spaces invite players
- Halo 4
 - feeling of approaching an epic otherworldly mythical presence



Any ideas where in human history we find something comparable?





Real World Architecture

- Design of Gothic structures
 - especially churches
- The Gothic Epoch
 - **end** of heavy fortification
 - lighter stretched **visual lines**
 - **look up** to see the end of the structure
 - **natural light** flood in emphasizing layering of columns
 - **focal point** on altar with triptychon telling a story for everyone to understand

“The whole Halo series is a masterpiece regarding how architectural structural elements can define game space.”





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06

Level Design Theory

Game Play Tools & Principles

Game Play Tools & Principles

- **Level Designers = Game Designers**
specialized in creating the space a player
experiences the world



- The world a Level Designer creates needs
to reflect the Game Designs intentions

“The world is put together from
the constructed restrictions and
rules to create the game for the
player.”

Game Play Tools & Principles

- Levels should have **three goals**

Adjustment of Behavior

Transmission of Meaning

Augmentation of Space

Game Play Tools & Principles: Adjustment of Behavior

- The primary function:

teach “**how to play** the game”
- **introduction** of new game **mechanics**
 - obstacles
 - repetition
 - players learn the mechanic and how to use it



Which games can you think of where the Adjustment of Behavior is happening? Give examples...

00:00:00

I-7: DIVERGE



Game Play Tools & Principles:

Adjustment of Behavior

- Hard to give a wrong answer e.g.:
 - Batman: Arkham Asylum
 - grapple and glide
 - Super Mario Bros.
 - different suites
 - God of War (2018)
 - first level
 - list continues almost endless
- **!! IMPORTANT FACTOR !!**

Game Play Tools & Principles: Transmission of meaning

- 2 important **factors** to **understand** a game
 - gameplay mechanics
 - narrative part
- good games use **both systems**
- interactive medium
 - cause and effect relationship
 - transported by Level Design
 - important storytelling part

“The level we are designing does **not always** have to play a **cutscene** or a movie element but can **convey the stress** or the “cinematic moment” **while playing**.

This is **more immersive** for the player than watching cutscenes passively, explaining the **consequences of their behavior**.”

JOURNEY

The image is the title screen of the video game Journey. It features a vast, surreal landscape with rolling pink sand dunes in the foreground. In the distance, a large, dark mountain peak rises above a layer of white mist. A bright sun or moon is positioned directly behind the peak, creating a strong lens flare and illuminating the scene. Two small, robed figures are walking away from the viewer up the slope of a dune on the right side of the frame. The sky is a pale, hazy green with soft, white clouds. The word "JOURNEY" is written in a large, white, serif font across the center of the image, partially overlapping the mountain and the sky.

Game Play Tools & Principles: Augmentation of Space

- **IT IS NOT** give more space and it will be better!
- create **specific areas** to **add** to the **experience**
 - e.g.: for boss encounters give a **breathing room** before the encounter
- create **patterns** with those areas
 - similar breathing rooms let players **anticipate** of what is coming
 - resulting in a **better understanding** of the situation = **better** gameplay **experience**
- adding in function to those areas
 - saving/loading points
 - loading resource heavy assets

Game Play Tools & Principles: Augmentation of Space

- build a **visual language**
 - which objects are interactable
 - breadcrumbs
 - gameplay loops
 - etc.
- use formal or spatial symbols for **player guidance**
 - players will follow and use established patterns
 - maybe we have adjusted their behaviour in that case? ;)

Game Play Tools & Principles

- three goals for a level

Adjustment of Behavior

Transmission of Meaning

Augmentation of Space

- turn into **important tools** for a Level Designer



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07

Level Design Glossar

Terminology

Macro Design

- “First step” to create a level
- design on the **large scale** focusing on the **overall space** of the level
 - visualize the **experience** as a whole
 - which **theme & landmarks**
 - which **gameplay** we want to integrate (requirements, playstyles, etc.)
 - which **emotions** we want to trigger
- **count of items** there need to be in the level, **determines the size** of the map
 - preventing squeezing things into place where there is none
- **IT PROVIDES THE BASE STRUCTURE** for a level
- **After** first Macro Design pass is done, **start building** the first “greyblock”
 - **reiterate** on Macro Design till we have **all elements** in place
- important to have **playtests** on Macro Design already

Greyblock

- **first in game version** of the macro design
 - usually with brush work & grey textures
- **creating first playable prototype** to test
 - timings
 - spacing
 - sightlines
 - technical pass
 - gameplay beats
 - gameplay loop usage
 - routing
 - logic
- **iterate** on greyblock till **core pillars** of the map are good
 - playtest with colleagues
 - get feedback
 - **evaluate and listen**

Micro Design

- **refine the Macro Design elements**
 - better **moment to moment** experience
 - react on feedback from greyblock
 - polishing
 - detailing
- **progress seems slower than on Macro Design**
 - more iterations turn the experience from ok -> good -> great -> etc.
 - variations if undecided
 - **focus testing** small changes

Gameplay Beat

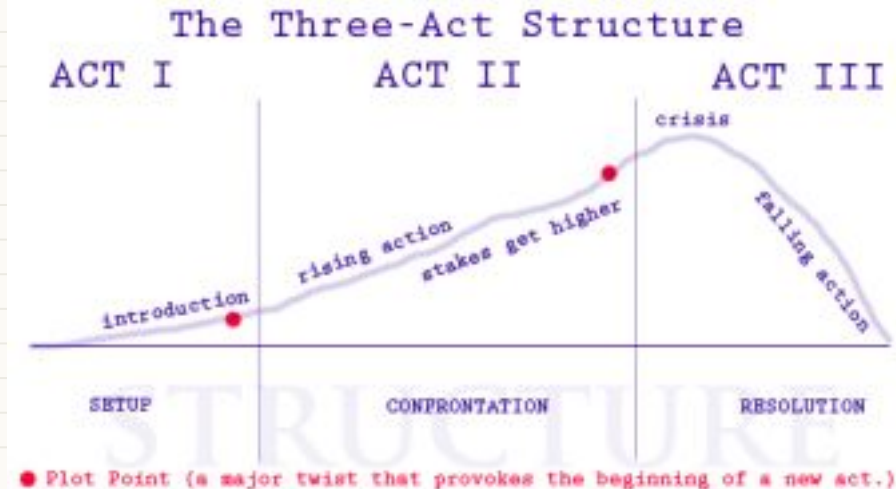
- an event and or activity in the level the player is going to perform
 - a level consists (most of the time) of multiple Game Play beats
 - Gameplay beats are defining the pacing of the map
- often separated into 2 categories
 - narrative beats
 - player based activities

Gameplay Loop

- **player interaction** with different **game systems**
 - e.g.: Player Unknowns Battleground has multiple gameplay loops
 - running and shooting
 - looting
 - inventory management
- 2 Gameplay Loops
 - core
 - secondary
- Gameplay Loops do use **different** amounts of **time** and **thought processes**
 - each loop should be **fun** and **rewarding** in itself
 - each loop should be **encouraging to repeat** itself
 - a **good mix** of gameplay loops creates **interesting gameplay pillars**
- do not mistake Gameplay Loops with Gameplay beats!
 - gameplay loops are designed by game design
 - gameplay loops are needed to resolve the gameplay beats utilized in Level Design

Excitement Graph

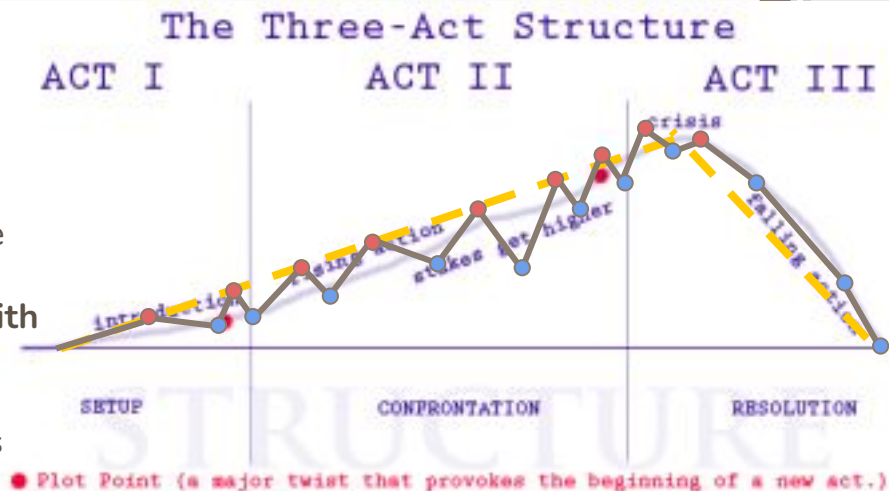
- overall flow/intensity/pacing how the **player experience** evolves through the level
 - e.g.: Three Act “Hero's Journey” as one example
 - Setup, Confrontation, Resolution
 - important Plot hooks



• <http://www.elementsofcinema.com/images/3-act.png>

Excitement Graph

- translation into a level
 - start in a **comfort zone**
 - build up need/drag towards an **objective** -> create a **goal** for the player
 - encounter** unfamiliar/dangerous situations
 - rise intensity throughout the level
 - create **breathing moments** for the player
 - let players **deal** mentally with what they **encountered**
 - creates emotion
 - intensity valley** emphasizes the next peak



<http://www.elementsofcinema.com/images/3-act.png>

Player Guidance

- systems of **visual language** to help players
 - pathing
 - telegraphing/signposting
 - constraints
 - gameplay loops/beats
- this is a **whole study**
 - like every **composition** needs to be designed
 - **decision** how much do I want to **hold the players hand**
 - which **established** patterns/gameplay functionality is there?
 - different for each game/level/situation
 - decision making with compromises

Player Anticipation

- give the player a **hint**/overview of what is coming
 - generate **drag** towards an **objective**
 - **prepare** for an upcoming boss battle
 - **establish** a visual language for
 - gameplay **loops**
 - gameplay **beats**
 - to let the players **learn what you want them to do**

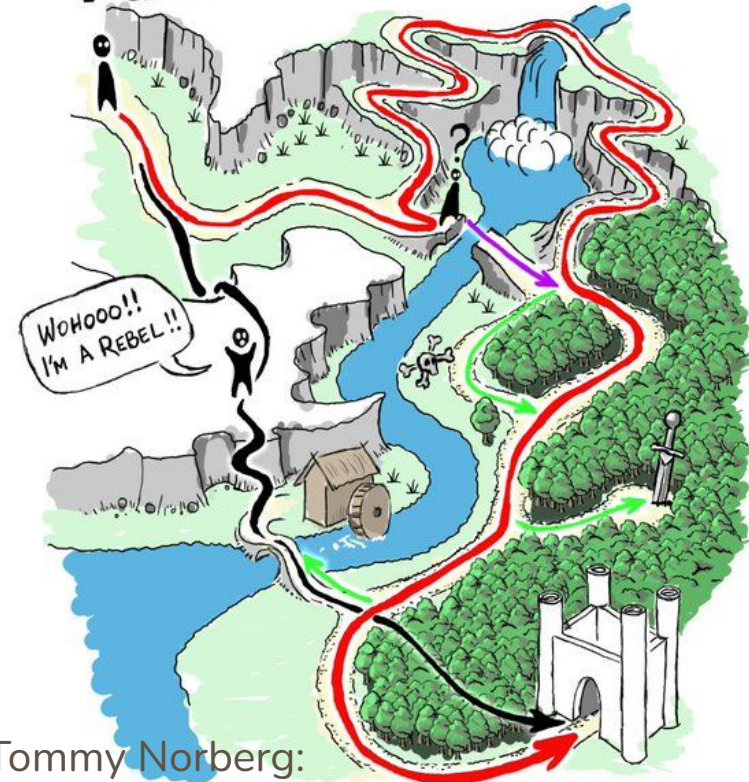
Readability

- how **clear** is the **situation** for the player
 - depending on **what** you want to **achieve**
 - find a **good balance** between **obvious** and completely **hidden**
 - sightlines
 - framing
 - cluttering
 - reduce visual noise
- goal is to have a **well composed scene** for the player to navigate with all design intentions in place
 - but the player should be **challenged** to solve a puzzle
 - let the player know about the puzzle
 - use appropriate guiding techniques

Paths

- There are **multiple categories** for paths throughout the level
 - Main Path:** (intended path, “golden” path)
 - The most obvious way to get from A to B
 - players should follow this route
 - we put it as the most obvious choice in our composition
 - Perceived Path:**
 - player anticipation points
 - look ahead what is coming
 - route actually takes you somewhere else
 - panoramic views

Paths!



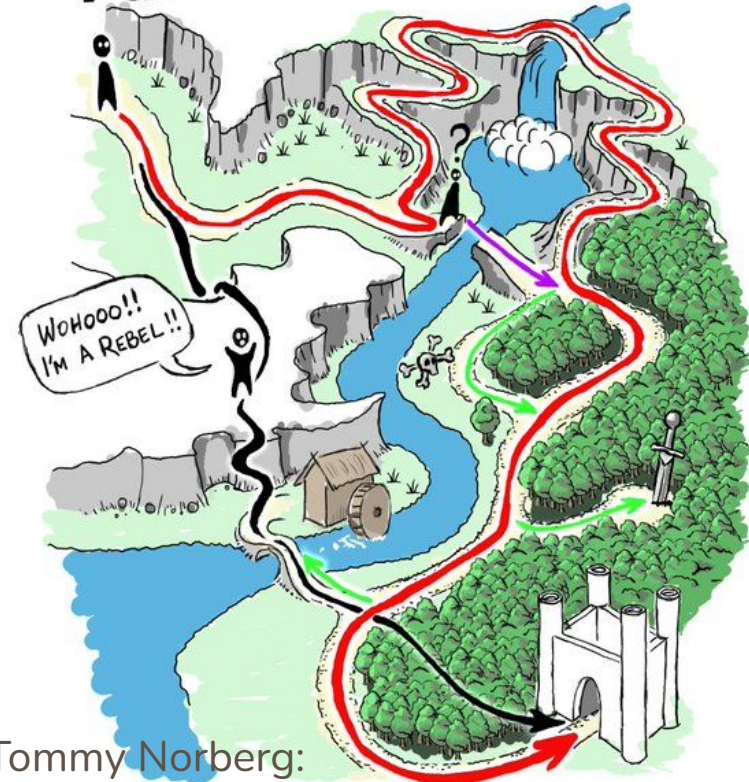
Tommy Norberg:

https://threadreaderapp.com/user/the_Norberg

Paths

- **Side paths/detours:**
 - alternative routing
 - find additional challenges/rewards
 - longer side paths should hold a meaning (sidequests etc.)
 - careful when you lead back a side path onto the main path
 - clear understanding where to go
- **Desire/Rebel path**
 - usually players will try to cut corners
 - let them do so
 - good to let them get onto the main path again on several points
 - but the more they cut the more they miss

Paths!



Tommy Norberg:

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Telegraphing/Signposting

- environmental assets helping with navigation
 - guide the players eyes with sight lines
 - poles along the roads connected with cables
 - bushes, treelines, etc.
 - road signs
 - where does the road lead



Constraints

- **blocking off areas**
 - river which can't be crossed
 - except over a bridge over there
 - moving along a mountain road
 - on side is a cliff with a nice view into the land we are going to
 - the other one is harsh unclimbable rock
 - an electric fence
 - the sea
- you will need to think of constraints for each situation
 - but **limiting** players on their movement should always give them **choice** on another side.



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08

Level Design Toolset

Game Engines & Editors

- many game engines and even more editors
 - Unreal & Unity are very good to learn
 - concept of a game engine
 - rapid prototypes
 - certain aspects not needed
 - expensive for release
 - most games have own tools to create levels
 - also have different needs

Where to start then?



Which tool does not change significantly?

Pen & Paper (or digital equivalent)

- established:
 - start gathering reference material on basic idea
- How to plan what the player should experience?
- How to plan for different gameplay beats?
- How do you tigh the gameplay loops together?
- Where to start and end the level?

“We start off with **clarifying** some general **questions** to set our **constraints**. The questions are universal and apply to all game levels.”

Pen & Paper (or digital equivalent)

- Which **audience**? Who is going to play the level?
 - ability to identify patterns
 - reaction times
 - attention span
- How **long** should the level be?
 - within the game universe
 - short medium long
- **When** is the level situated in the game?
 - player progression & established patterns
 - complexity
 - excitement graph
- **What** do I want/need to integrate/SHOW OFF?
 - gameplay loops & beats
 - story elements
 - specific tech elements

Clarify the most critical constraints!

Story, theme and plot for the level are the most important constraints you have to follow. What are my **building blocks**? Do I have **metrics** to follow? You can derive this from the other constraints given.

Pen & Paper (or digital equivalent)

- example game:
 - Narrative driven adventure game
 - about a person living on it's own
 - the partner recently died (in an heroic act? -> firefighter?)
 - the player plays this character
 - getting through the day to day struggle
 - and living through the 5 stages of grief
 - denial
 - anger
 - bargaining
 - depression
 - acceptance
- I am not a storywriter! so this is just to set a scene!

Pen & Paper (or digital equivalent)

- example game continue
 - **narrative elements** are
 - the house with its interior
 - the garden
 - house change based on player progress to reflect current stage of grief
- **Level 1 - Introduction**
 - introduction cutscene - sets scene
 - what has happened
 - character gets out of bed, calendar with back to work
 - washes, puts on clothes from pile of unfolded t-shirts
 - walks down the stairs, house is a mess
 - leaves the house, door closes
 - player takes control
 -



Pen & Paper (or digital equivalent)

Audience?	Curious Students, good at identifying patterns, have a good reaction time, attention span (?)
How long?	short to medium, in comparison to later interactions with the house short
Player progression	none, introduction level
Complexity	introduction, easy, learn basic mechanics of the game
Excitement Graph	ease in, flat curve with slight peaks, 3 acts + resolution
Gameplay	movement, quest system, puzzle element, object interaction
Story	Char needs to go to work after 2 weeks since the partner died. house is a mess, the garden is well kept no key for the car, rain (instrument of feeling) doesn't allow to walk to office, front door locked, char remembers partner always kept the rear door open. -> enters house walks through rooms - lots of memories solving puzzle-> finds keys -> can start car -> cut scene
Tech	Weather system, movement, quest system, object interaction

Pen & Paper - Macro Design

- Macro Design
 - visual representation of answered questions
- how to get there is personal preference
- general schematic of the level structure
 - Start/End Point
 - what needs to be integrated
 - fulfill purpose of the level
 - pathing ideas



What do we do with this?



Evaluate and get feedback!

Pen & Paper - Macro Design

Start

Movement

Garden

Story beat 1

Quest
System

House

Story beat 2

Object
interaction

Car

Story beat 3

Puzzle
element

weather
system

End

Pen & Paper - Macro Design

Start

House

Garden

Car

Garden

House

Movement

Quest
System

Object
interaction

Quest
System

Puzzle
element

Story beat 1

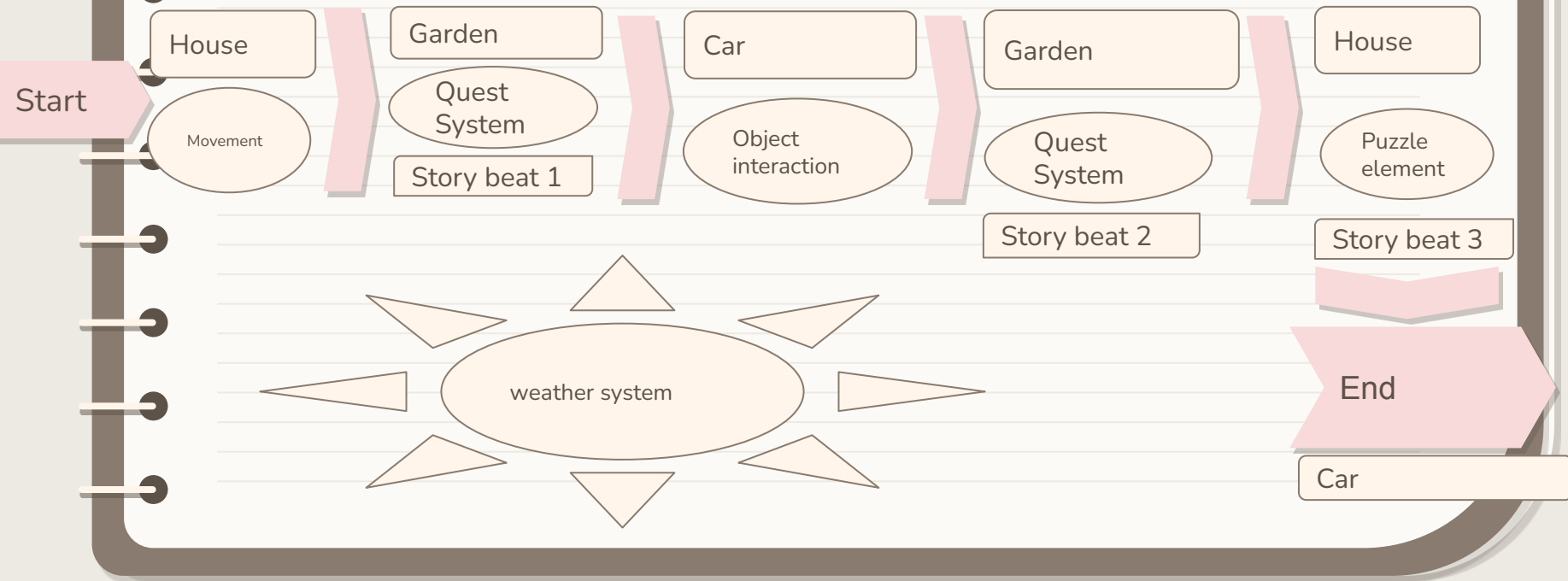
Story beat 2

Story beat 3

weather system

End

Car



Pen & Paper - Macro Design

- Macro Design
 - visual representation of answered questions
- how to get there is personal preference
- general schematic of the level structure
 - Start/End Point
 - what needs to be integrated
 - fulfill purpose of the level
 - pathing ideas
- getting to a **bubble diagram** / flowchart
- we did the **first description** of the level

Pen & Paper - Macro Design

- **iterate** on the schematic and **find flaws** with in its logic
 - Have you integrated all game play beats?
 - Is the flow of the map correct?
 - Where is room for improvement?
- What is needed to **realize** the actions in **the bubbles**?
 - the amount of items in a bubble are also a good indicator of the importance in the level
 - how much space do they need
- transforming the schematic into a **plan of actions**
 - **detailing** the needs for **each bubble** from the players perspective
 - the **bigger** a **bubble** gets the **more space & importance** it has
 - do a clean drawing -> start **"Paper Design"**
- get feedback on it

Pen & Paper - Macro Design

- **Budget enough time for Macro Design**
 - **good planning** in the beginning **saves time** on the long run
 - limiting room for error/rework
 - important in the industry
 - time is money
 - more important
 - **health (mental and physical)**
 - do not burn yourself out and also not others!
 - do not crunch 70 hours a week for months
 - when this is happening in a job
- there is a lot going wrong in the planning phases -



Lecture
Overview

Introduction

History of
Game Spaces

Essential
Skills

Level Design
Theory



Questions?

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Level Design Lecture

FH Salzburg

Thomas Bauböck WS 2025 - Unit 4



09

Single Player Level Design

From Macro to Micro

Singleplayer Map classification

- three major categories
 - linear level
 - Multi- Path level
 - “open world”/“sandbox” levels
- all viable design approaches in the right place
 - **complexity** for players
 - **game constraints**

Start

Path

End

Start

Path 1

Path 2

Path 3

End

Sta

End

Start

End

End

Path 1

Path 2

Start

Start

End

Start

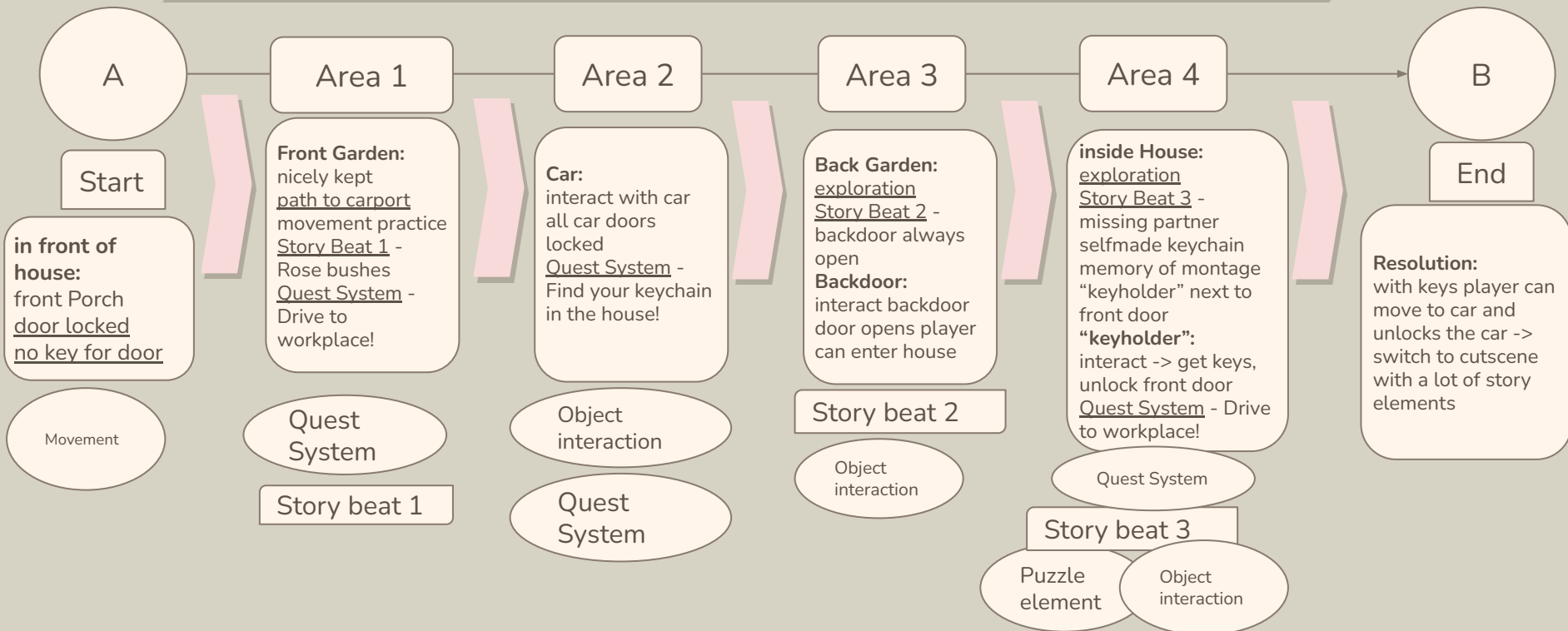
Linear level

- **linear level** brings a player from A to B
 - connected by predefined path
 - [A] Start Point / [B] Endpoint
 - macro design
 - encounter X gameplay elements along the way



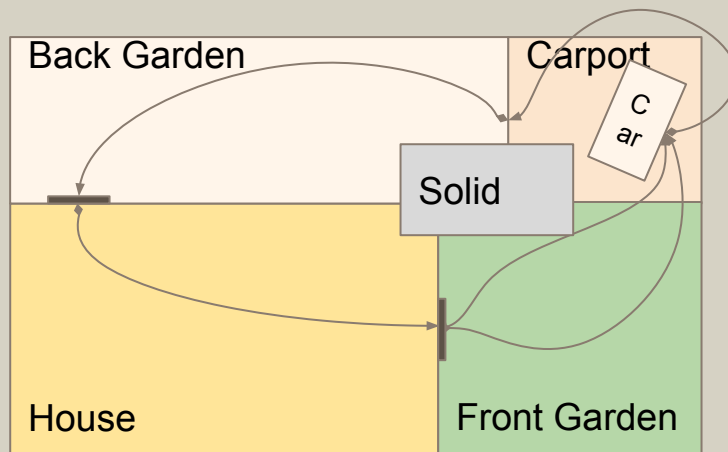
- Each “**Gameplay beat**” needs their **own space** to be memorable enough for the player.

Linear level - schematic/bubble graph



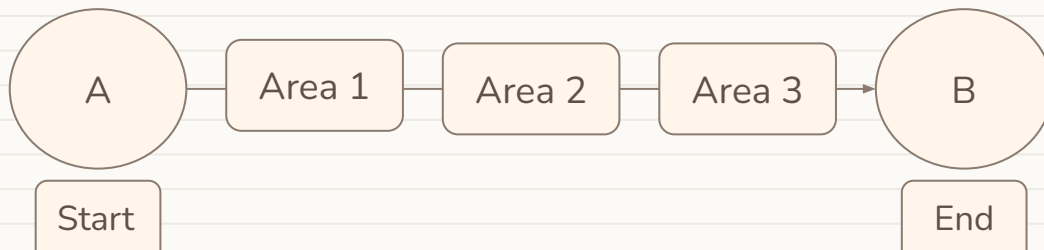
Linear level - from schematic to layout

- design a **layout**, following our **flow** of **gameplay beats**, according to the excitement graph
- creating “areas” (gameplay beats)
- and connectors (breathing moments)
 - decision making where to put things



Linear level

- Designers know **where** the player is for each gameplay beat
- Designers have full control on **when** the player encounters **what** on the path
- **full control over the experience!**



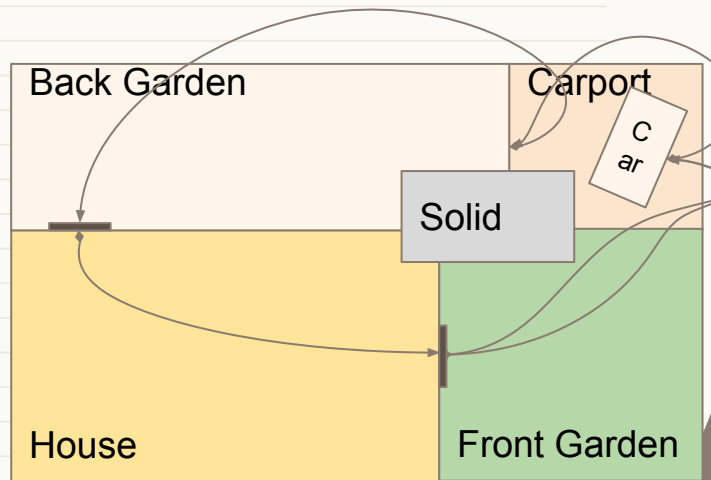
Linear level



When are linear Level Designs used?

- Tutorial Levels
- Preparation for a Boss battle
- focus on gameplay / narrative
- structure the core route
 - basic idea is easier to sell to stakeholders
 - mention if you plan to expand on the design to a more complex level

All this due to general layout simplicity for the player to understand.



Multi Path Level

- explore the world they are in
- find their own path
- replayability
- player decision

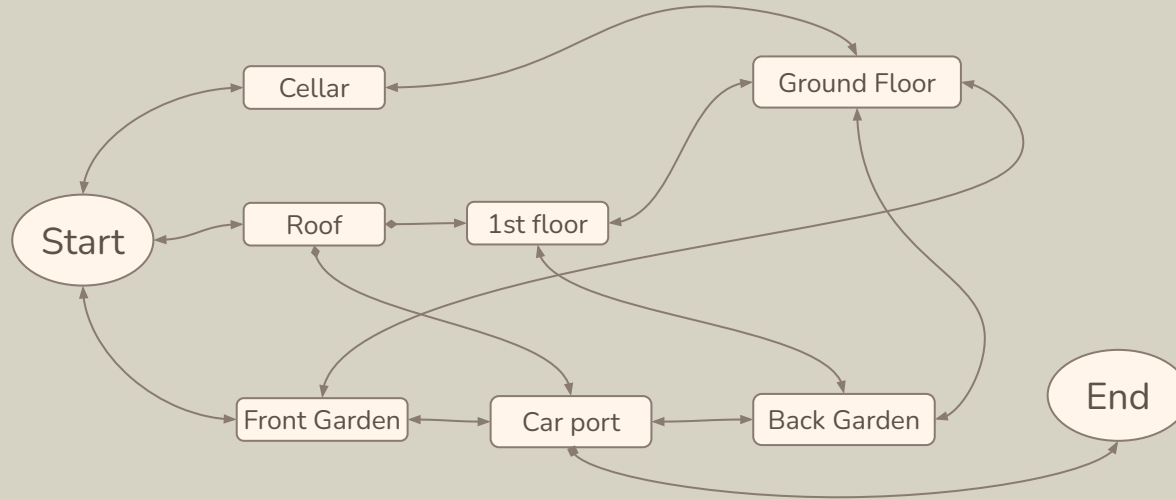


How are we going to plan a
“Multi Path level”?

A Multi path level is designed to facilitate **multiple routes** through an environment with **intertwining paths**, leading from the **start** to a defined **end** point. The player is given **CHOICE** on how to achieve this goal!

Multi Path Level

- Start / End point
- adding more paths
- crossing each other in several junctions



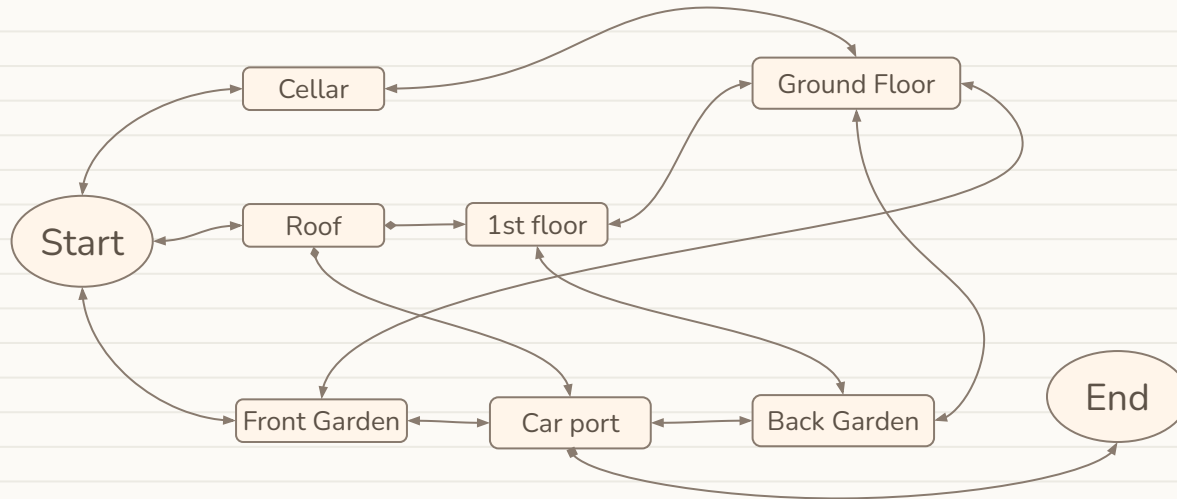
It's easier for humans to **decide between two** options than between three.

Three options are mostly seen as a **luck based** outcome.

The decision between **two** are a **more meaningful** decisions and therefore establish a **stronger emotion**.

A **three way** decision can be used if the player has **not too much information** on the matter already. Mostly used in the **beginning of a level** or as a routing solving junction

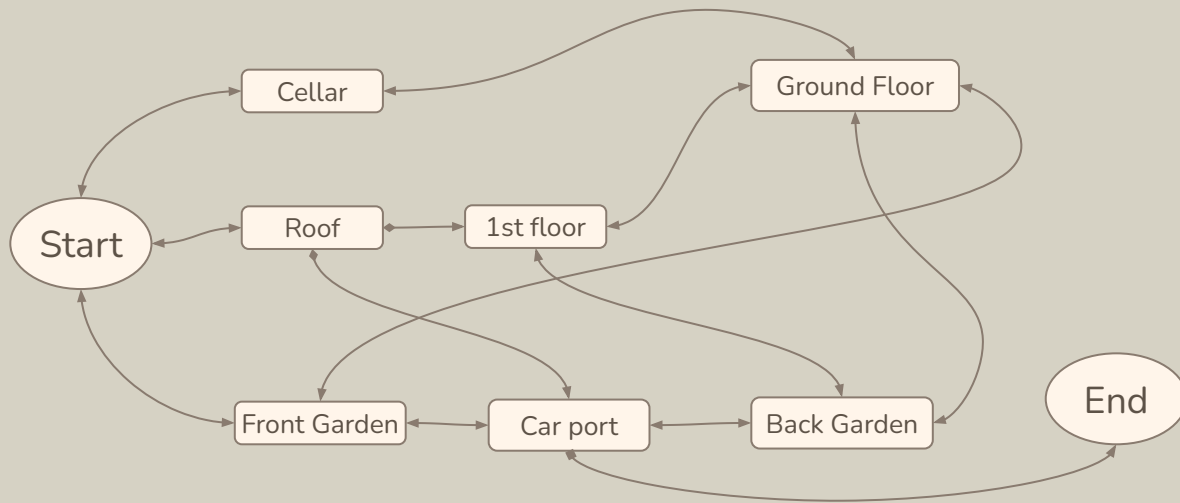
Multi Path Level



Which problems as a Designer are we encountering here?

Multi Path Level

- routes miss or repeat gameplay beats
- finish map without hitting all beats
- more prone to backtracking
- skewed/bad excitement graph
- bad experience overall



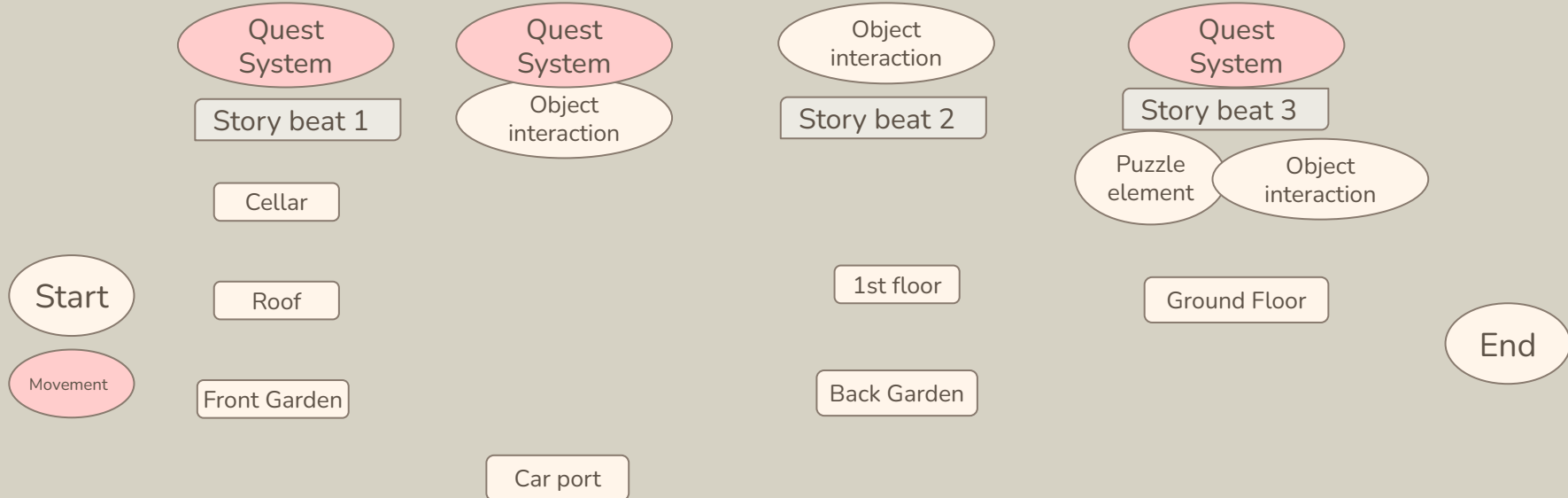
Multi Path Levels are getting more complicated very quickly.

Analyse and break the problem **down** into manageable pieces.

Evaluate again after adjustments!

Multi Path Level

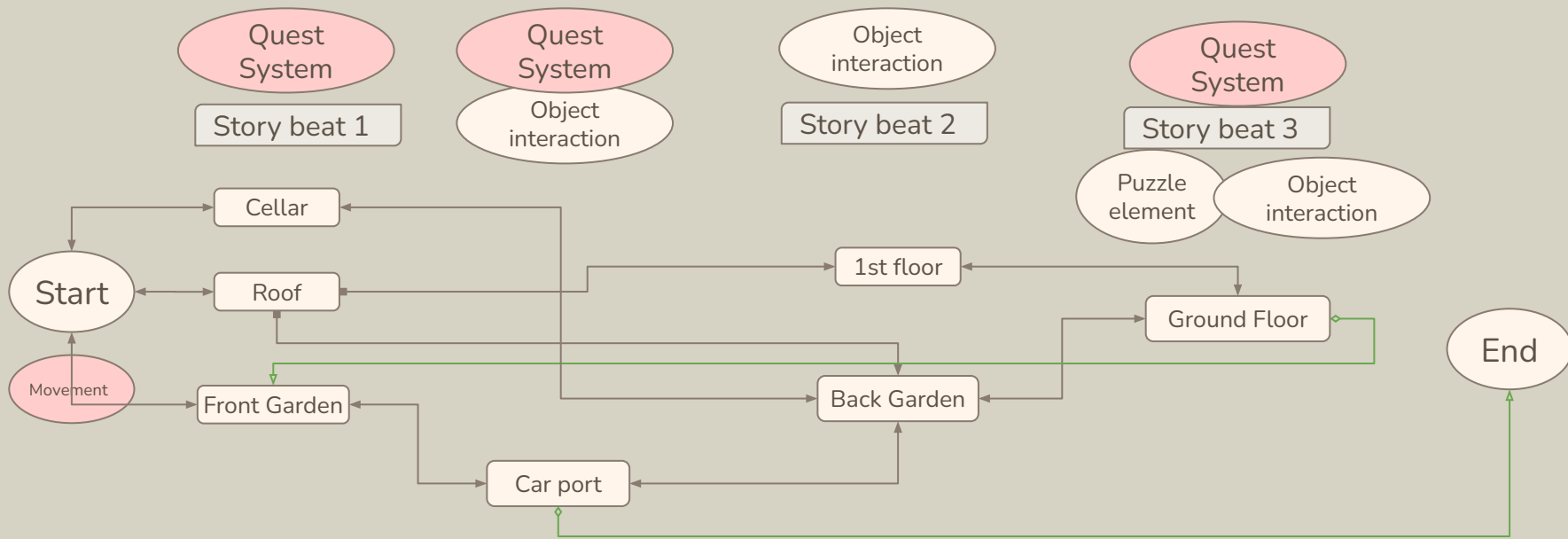
- Problem 1: Gameplay beats
 - assign locations to certain gameplay beats
 - align them in the flow graph / schematic
 - knowing where player is for each beat



Multi Path Level

• Problem 2: Pathing

- each route should hit each beat at least once
- adjust alignment of beats and routes
- plan junctions for player choice
- player guidance is crucial to push & pull into the next beat instead of backtracking



Multi Path Level

- Room for error
- designer sacrifices control over player to achieve
 - player driven
 - gameplay oriented
 - “richer” experience

Multi Path Levels are getting more complicated very quickly.

Analyse and **break** the problem **down** into manageable pieces.

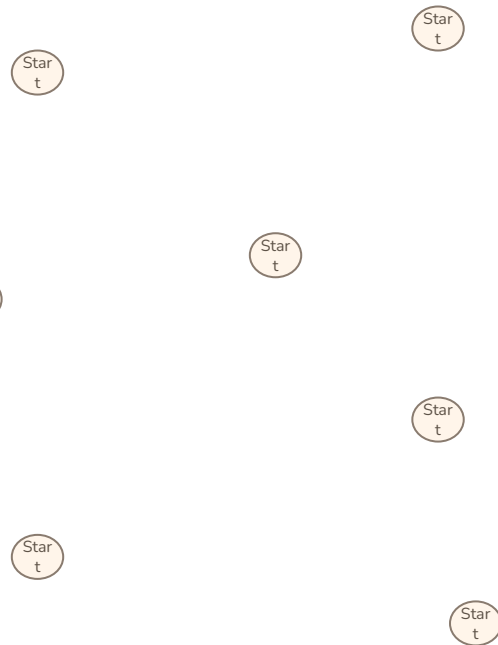
Evaluate again after adjustments!

“Open World” Sandbox levels

- importance of **player location** in correlation of **game beats** and **excitement graph**
- in “Open World” levels the **player is free** to go where they want (or are they not?)
- “Open World” levels need to **constrain the player**
 - constraints the player **does not notice**
 - constraints the player **thinks it** was their **decision**, they have been **clever enough**
 - Subtle **pushing and pulling** towards the objectives we want the player to experience
- there is **no defined start point**
- the player can be everywhere



Any ideas what we should do to still plan our level?



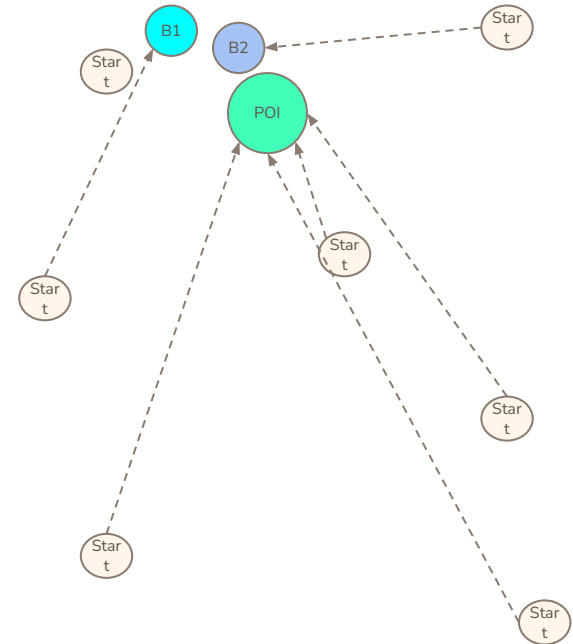
“Open World” Sandbox levels

- simplify the problem
- instead of all possible start points we pick one
 - for the other spawn points, we deal with them later

Start

“Open World” Sandbox levels

- simplify the problem
- instead of all possible start points we pick one
 - for the other spawn points we deal with them later
- and we put an end point for our level a “Point of Interest” - POI
- design a path towards the objective according to an excitement graph
- we do have several gameplay beats in the map
- one spawn point taken care of
- all other spawn points also have a POI
- with different paths of course

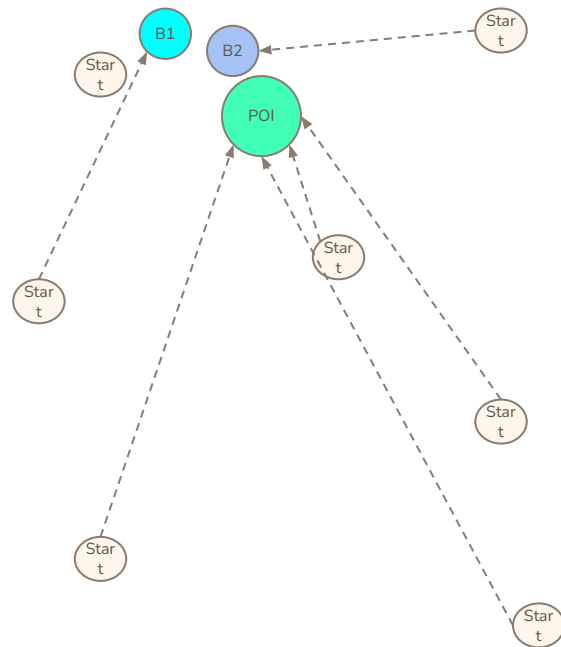


“Open World” Sandbox levels

Players are **PLAYING** a game.
It is in our heart to have a
certain amount of **curiosity** to
find out what is behind the
next corner or what this
abnormal tower sitting on the
cliff over there is all about.
Players will find a way.

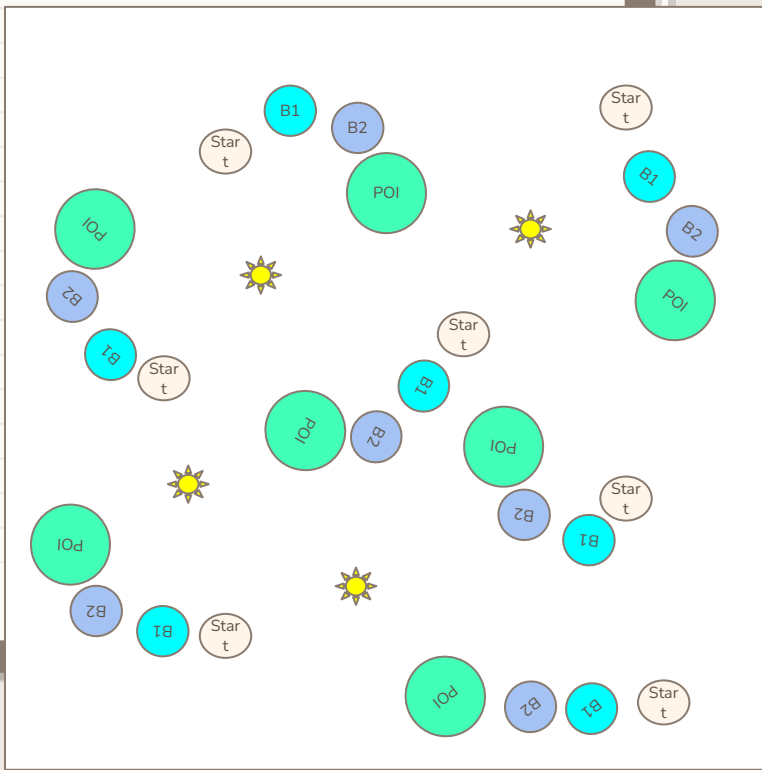
“Open World” Sandbox levels

- the other paths are closer / further away and according to our excitement graph will give different experiences
- Distance & travel times most impactful on decisions
- also
 - importance
 - different
 - valuable
- How much do I know about it?
- Has it already been visited?
- Gameplay elements like
 - “Do I have to kill the big Red Dragon in that cave?
 - And how many dots do I need to kill it?”



“Open World” Sandbox levels

- use those variables to create more Pols
 - create an environment where the player has choices where to go
- Exploration is a good excitement graph on its own
 - put little rewards off the beaten paths to emphasize this
- actual gameplay beats hard to sort in a flow chart
- granulate them into smaller steps in between
- use gameplay mechanics:
 - quest chains with traveling
 - adapting difficulty/reward systems
 - etc.

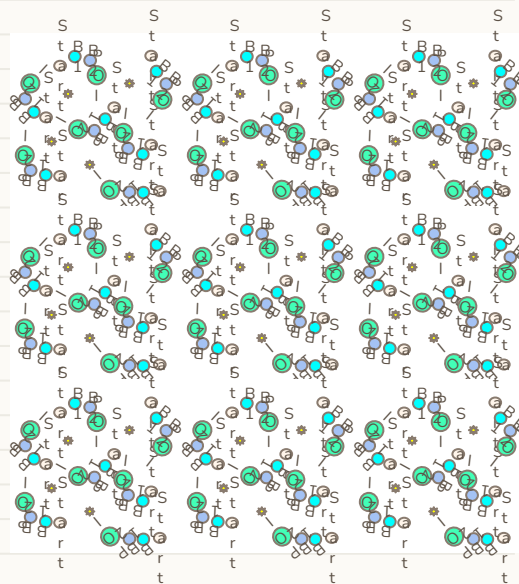


“Open World” Sandbox levels

- Open World games consist of more than 7 Pols
 - difficult to keep player on “the path”
 - side/backtracking is unavoidable
- Simplify the problem: What do we want?
 - experience a collection of Pols



How does the player know about the collection of Pols?



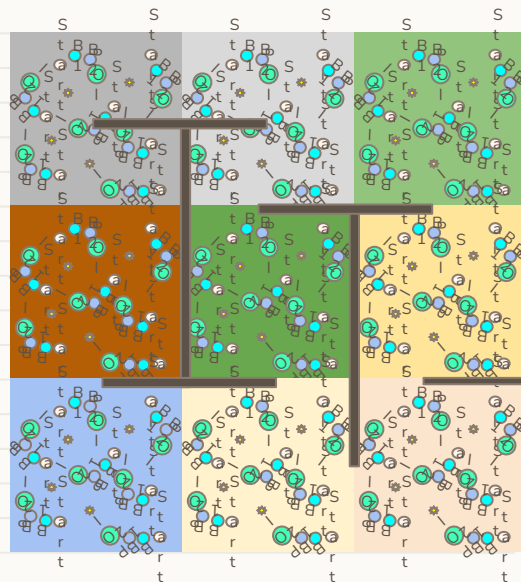
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- Open World games consist of more than 7 Pols
 - difficult to keep player on “the path”
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- Simplify the problem: What do we want?
 - experience a collection of Pols



How does the player know about the collection of Pols?

- Theme them and create a region for the player
- build a border to set a constraint and limit player movement -> biomes / regions
- not 100% ensure the player will follow the path
 - but he is highly incentivized





<https://guided.news/wp-content/uploads/2022/09/GTA-5-Fan-Map.jpg>

https://upload.wikimedia.org/wikipedia/commons/3/31/Kalimdor_english.svg

“Open World” Sandbox levels

- **segment** “Open World” into regions
- regions can be **further segmented** into smaller regions
- **layered structure** - bubble by bubble breaking down the region into **smaller pieces** till we are at a
- **Multi Path level** entering a house from whichever direction we want
- going down into the cellar through one entrance
- **A linear level**
 - to simplify navigation
 - where the gameplay beats are fully controllable
 - players pick up some conveniently placed items
 - and encounter the end boss

The **layered structure**, getting **simpler navigation** the further down you go in the onion shells, **prevents** overly **complex situations** for the player.

The **simpler navigation** the **more complex** other **gameplay loops** can be.



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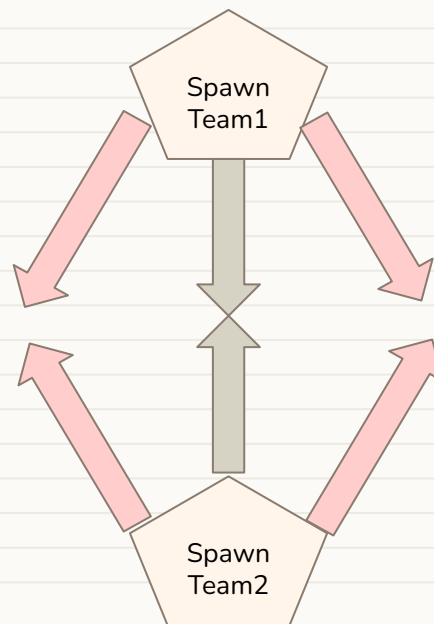
Multi Player Level Design

Three Lane Map Design

- 2 teams fighting each other
- many genres due to versatile adaptation
- components of 3 lane maps
 - 2 team spawns - Team 1 & Team 2
 - 1 main route
 - 2 side routes
- easy to create loads of different maps
 - variety in gameplay loops
 - different gameplay beats
 - interconnectivity

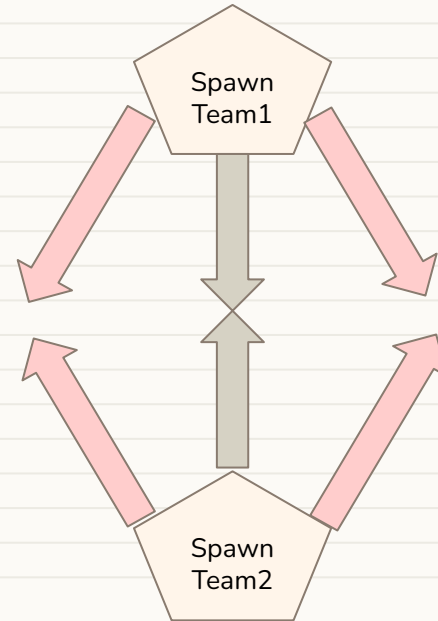


Why is it so good?



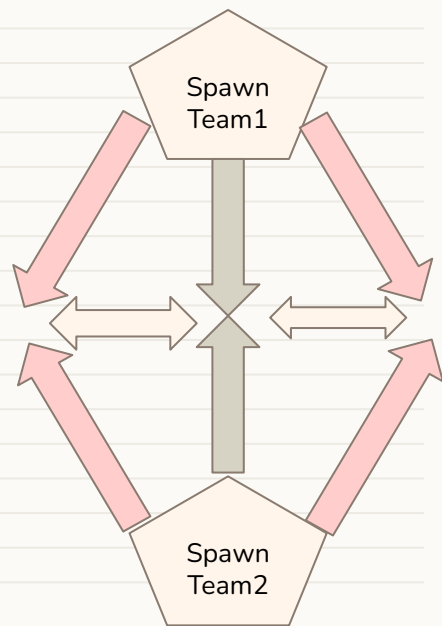
Three Lane Map Design

- Subitizing
 - ability to assess an amount of objects at a glance
 - humans usually subitize 3-5 objects
 - sometimes 6 or 7 but more likely
 - subitize 2x3 or 3+4
 - with 8+ it is definitely counting or subitizing pairs and patterns
- Humans can asses 3-5 things at once.
- Three lane map design
 - 2 spawns
 - 3 lanes
 - 5 sectors of the map



Three Lane Map Design

- at the beginning the player can choose between 3 options
 - a good amount for initial player decision
 - easy to memorize with anticipation phase
- each lane has 5 sectors
 - Spawn, route to clashing area, clashing area; route to enemy spawn, enemy Spawn
 - easy to learn the route and memorize it
 - easy routing -> better for complex gameplay loops
- interconnectors where to go?
 - easy decision making en route
 - center lane need a “clever” solution to have this

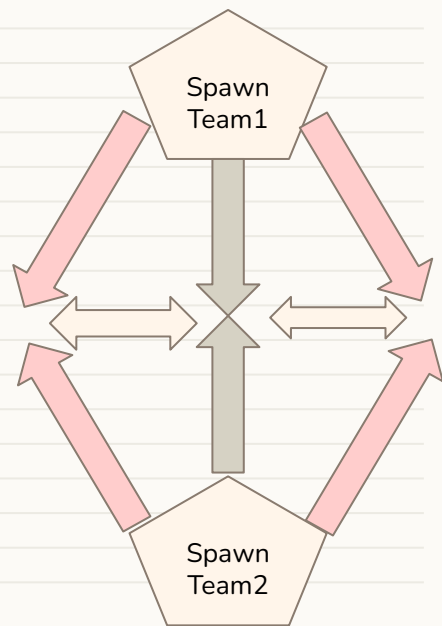


Three Lane Map Design

- Three Lane Maps easy to navigate
 - easy to navigate -> complexity in gameplay beats
- Focus in competitive games on gameplay beats
 - individual skill levels of players determine team outcome
 - players should compete against each other
 - NOT against the map

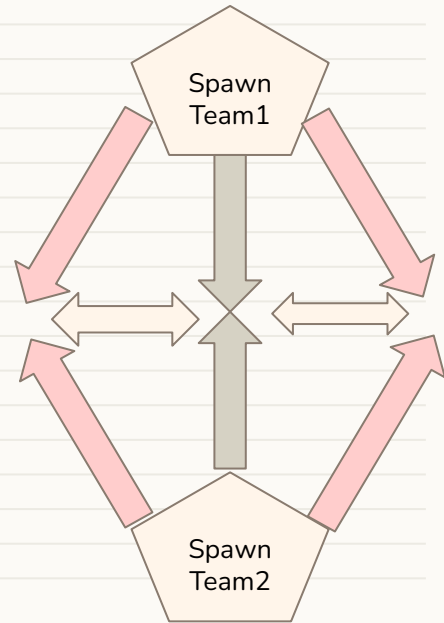
Three Lane Map Design

- The central lane or “middle”
 - usually shortest connection from spawns
-  What are the Implications for the shortest route?
- players are more likely to take the middle
 - distance motivator
 - quicker in action
 - more enemies
 - often has connectors to sidelanes to retreat into when mistakes have been made
 - retreat option



Three Lane Map Design

- The side lanes or “flanks”
 - flanks have one side less to think about
 - less routing complexity
 - more gameplay complexity
 - variety of options to engage the enemy
 - Usually contain a **Combat Focal Points**
 - The area of a map in which the decisive forces of a combat meet.
 - get behind enemy lines in middle



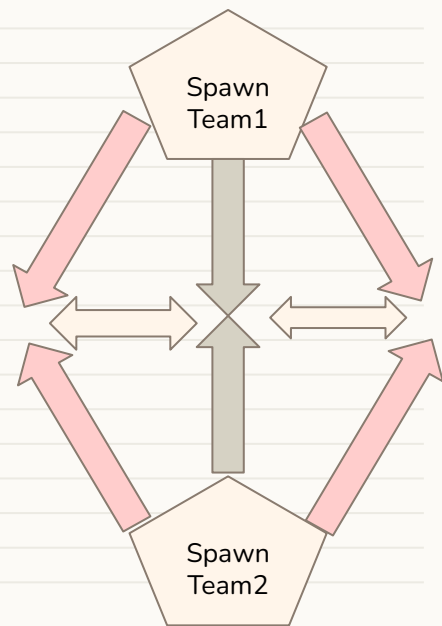
Three Lane Map Design

- Timings

- short middle
- longer side lanes
- connectors

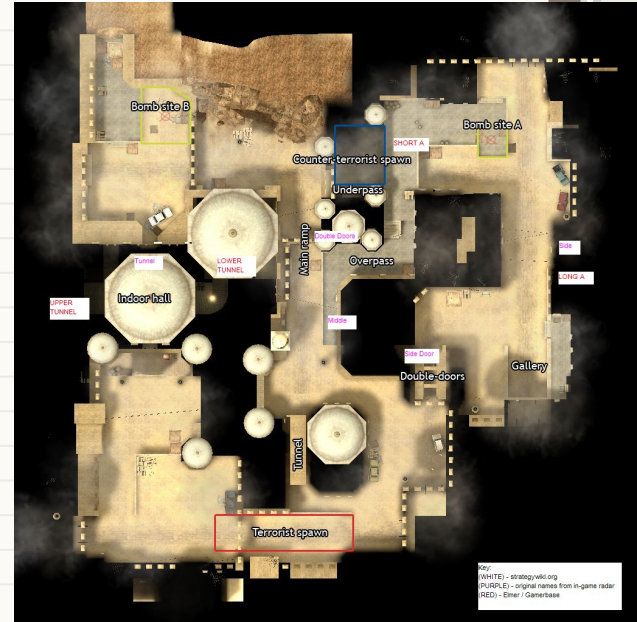


How do we get to a well distributed player concentration over the map?



Three Lane Map Design

- Central lanes use gameplay mechanics to slow down the players
 - choke point which can be controlled from a far



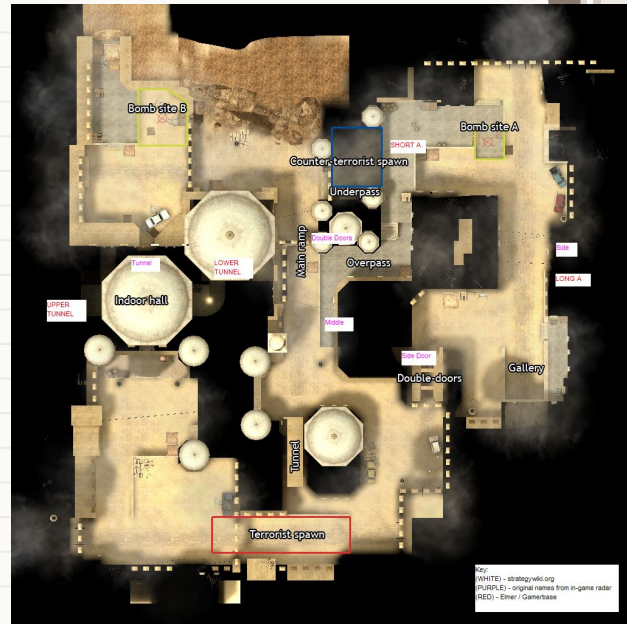
Three Lane Map Design

- Central lanes use gameplay mechanics to slow down the players
 - choke point which can be controlled from a far
 - open spaces with sparse cover elements
 - gameplay for slower classes



Three Lane Map Design

- Side lanes
 - more precise/coordinated gameplay
 - higher skill requirements to outplay enemy
 - dynamic gameplay elements to move around, getting deeper into enemy territory



Symmetrical Level Design

- easy to learn for players as both teams have
 - similar cover setup
 - routing & timing
 - areas of engagement
- easy to balance for same reasons
 - outcome mostly based on player skill
- multiple symmetrical maps in same game mode feel alike
 - waters down uniqueness and replayability
 - gameplay needs to be very strong and diverse to balance out
- **STILL VERY VALID MAP STRUCTURE!!!**
- MOBAS, Capture the Flag, Class based Team shooters



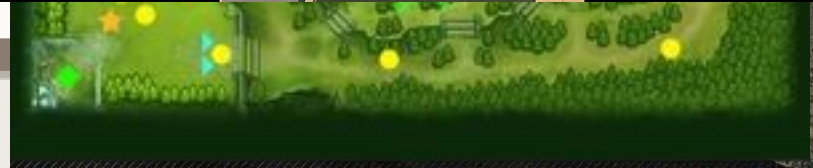
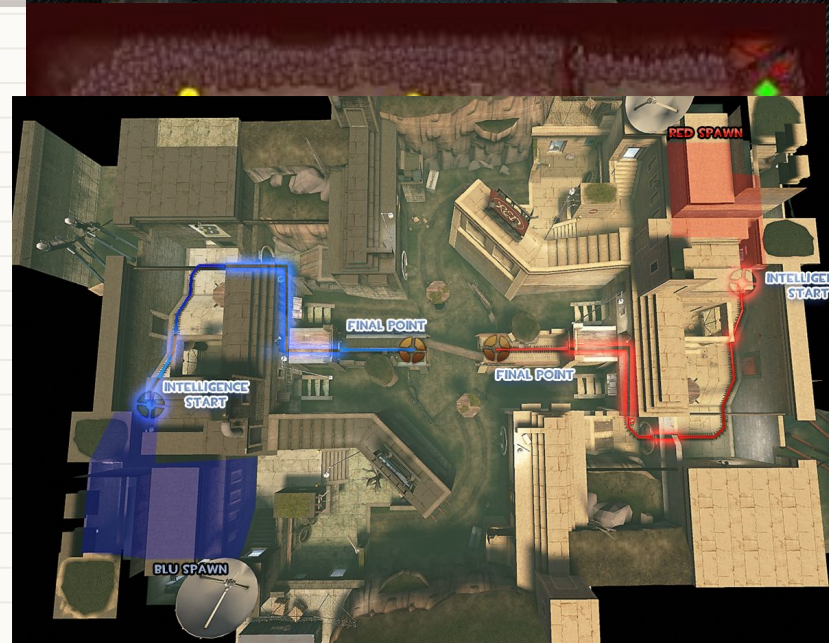
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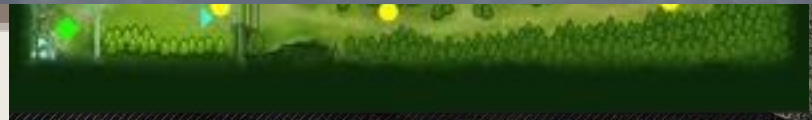
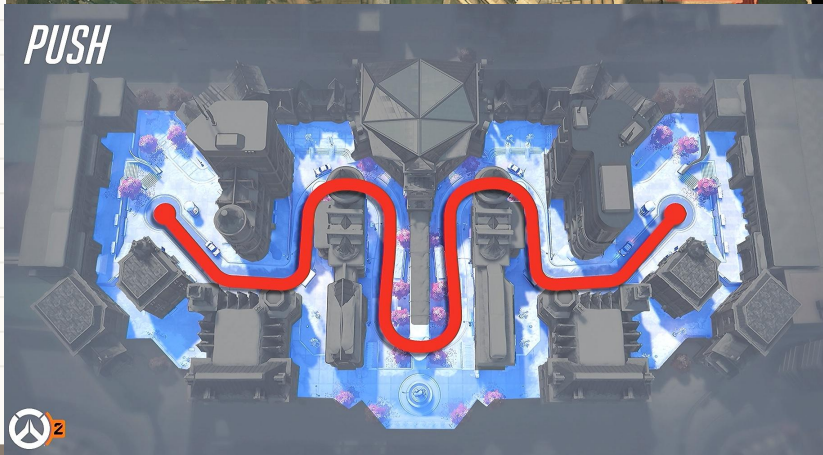
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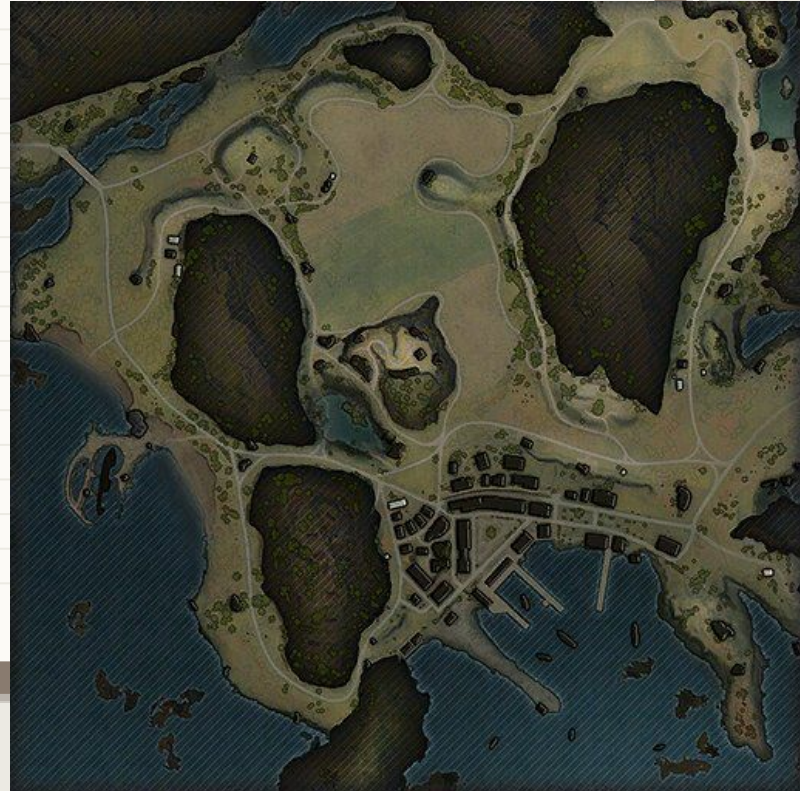
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Asymmetrical Level Design

- provides a more interesting layout
- creates more tension while playing
 - traverse more unpredictable gaming space
 - alteration of gameplay patterns
 - more natural realistic look and feel
- supports more diverse game modes
 - 2 teams do not have the same goal
- harder to balance



Asymmetrical Level Design

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 - traverse more unpredictable gaming space
 - alteration of gameplay patterns
 - more natural realistic look and feel
- supports more diverse game modes
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DE_INFERNO

- 1 T-SPAWN
- 2 CT-SPAWN
- 3 A-SPOT
- 4 B-SPOT
- 5 OBSTKISTE
- 6 Y-ECK
- 7 DUNKLES-ECK/DARK
- 8 BRUNNEN
- 9 KABELTROMMEL/KABEL
- 10 BEET
- 11 SANDSÄCKE
- 12 RUINE
- 13 GELB/POOL/HOF
- 14 LKW
- 15 HUNDEGANG/CT-GANG
- 16 SAKURA
- 17 EMULATE/SANDSÄCKE
- 18 TAXI
- 19 BANANE
- 20 HOLZ
- 21 T-RAMPE/T-SCHRÄGE
- 22 FUCHSBAU/ARSCHGANG



- 23 AUTO/FERRARI
- 24 MITTE/CT-MITTE
- 25 BANK
- 26 T-BRÜCKE
- 27 T-HAUS
- 28 FALSCH MITTE/T-HAUS
- 29 HEUWAGEN
- 30 KINDERZIMMER
- 31 DÖNER/SCHLAFZIMMER
- 32 HAUSMEISTER
- 33 LAUBE
- 34 TEPPICH/TEPPICHGANG
- 35 ACTION/KURZ
- 36 BALKON
- 37 LKW
- 38 GROßER SANDKASTEN
- 39 FRIEDHOF
- 40 KLEINER SANDKASTEN
- 41 BIBLIOTHEK/BIBO
- 42 HAZZ/LANG
- 43 CT-BOGEN
- 44 ARSCHECK
- 45 RIPPEN



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Thank You!

End of unit 4!

bauboeckthomas@gmail.com





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Appendix

Appendix

Trello LD COMPENDIUM:

a collection of articles all around the topic level and GameDesign

<https://trello.com/b/AM3ZOmAd/level-design-compendium>

Tommy Norberg: https://threadreaderapp.com/user/the_Norberg

Iulio-Cosmin Oniscu: <https://iuliu-cosmin-oniscu.medium.com/>

